Trade in AI-Related Products

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The views expressed herein are those of the author and not necessarily those of the Federal Reserve Bank of Minneapolis or the Federal Reserve System.

The AI Infrastructure Boom and Trade

The U.S. is experiencing a massive AI-investment-driven boom — trillions of dollars invested in data centers and related AI infrastructure.

This paper asks: How is the AI boom influencing international trade flows?

Trade in AI-related products is *the* dominant force in U.S. trade flows over the past two years. Not changes in U.S. trade policy.

Six questions...

- How to classify AI-related products?
- What are AI-related products?
- How much trade in AI-related products?
- Where are these products coming from?
- How has trade policy treated AI-related products?
- Does this matter for the aggregate?

How to Classify AI-Related Products?

Use an LLM to classify each of the ∼18,000 HS10 commodity codes by relevance to AI data center infrastructure. Snippet of the prompt:

```
You are an expert in AI data center construction, operations, and
supply chains.  Your task is to classify products by their relevance to
building and operating AI-focused data centers (hyperscale facilities
running GPU clusters for training and inference).

Consider these categories of relevant products:
- COMPUTE: GPUs, CPUs, memory, PCBs, servers, storage drives
- NETWORKING: Fiber optics, switches, routers, transceivers
- COOLING: Chillers, cooling towers, CRAH units, fans, pumps
- ELECTRICAL: Transformers, switchgear, UPS, batteries, generators
- BUILDING: Structural steel, concrete, rebar, raised floors
- SPECIALTY MATERIALS: Rare earths, copper, aluminum...
```

Each HS10 code gets: **Relevance**, **Category**, **Confidence**, and **Reasoning**.

How to Classify AI-Related Products? — An Example

The LLM sees the HS10 description *and* the NAICS industry code:

```
HS10:  8542310040
Description:  PROCESSORS (INCLUDING MICROPROCESSORS): GRAPHICS
PROCESSING UNITS (GPUS)
NAICS: 334413 --- SEMICONDUCTOR AND RELATED DEVICE MANUFACTURING

RELEVANCE: High   CONFIDENCE: 100%
CATEGORY: Compute Hardware
REASONING: GPUs are the core compute hardware in AI data centers,
specifically designed for parallel processing required in ML training
and inference.
```

Baseline analysis focuses on products classified as **High** — those directly used in data center construction or operation.

Result: 645 High-relevance codes out of 18364 total.

What Are AI-Related Products?

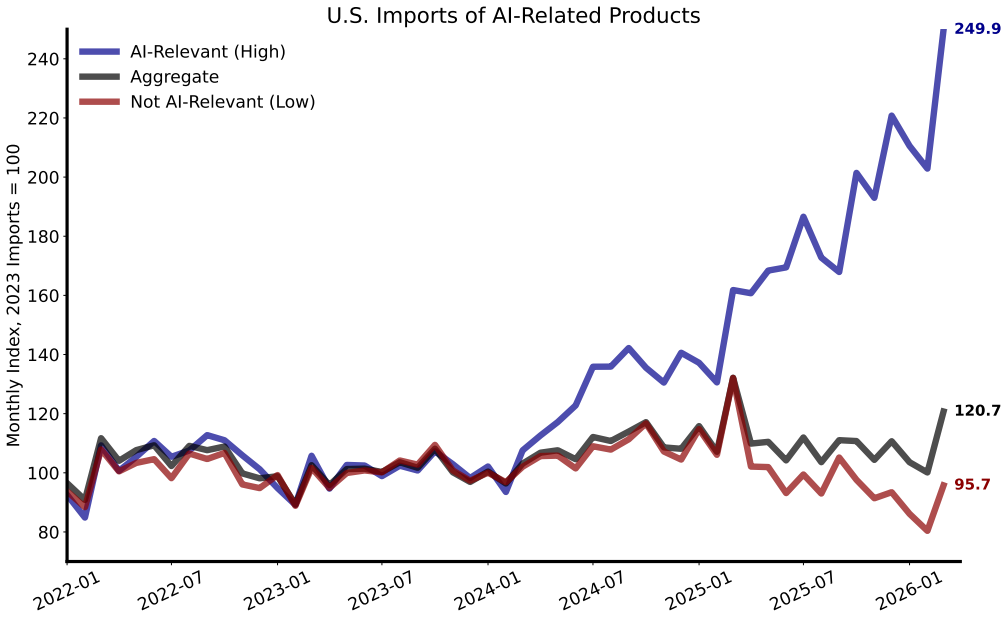
Category	# HS10 Codes	2023 ($B)	2025 ($B)	Change (%)
High AI Relevance	**645**	**379.0**	**654.0**	**+72.6**
Compute Hardware	163	144.4	353.8	+144.9
Electrical Power	250	116.9	141.8	+21.3
Networking Telecom	24	62.9	99.5	+58.2
Cooling HVAC	137	41.5	47.5	+14.4
Building Structure	44	12.1	10.1	-16.5
Fire Safety Security	8	0.6	0.7	+10.5
Specialty Materials	19	0.4	0.5	+22.5
Low AI Relevance	15,915	1,834.7	1,880.9	+2.5
Total Trade	**18,364**	**2,598.3**	**2,883.9**	**+11.0**

What Are AI-Related Products? — Top 10 by Import Volume

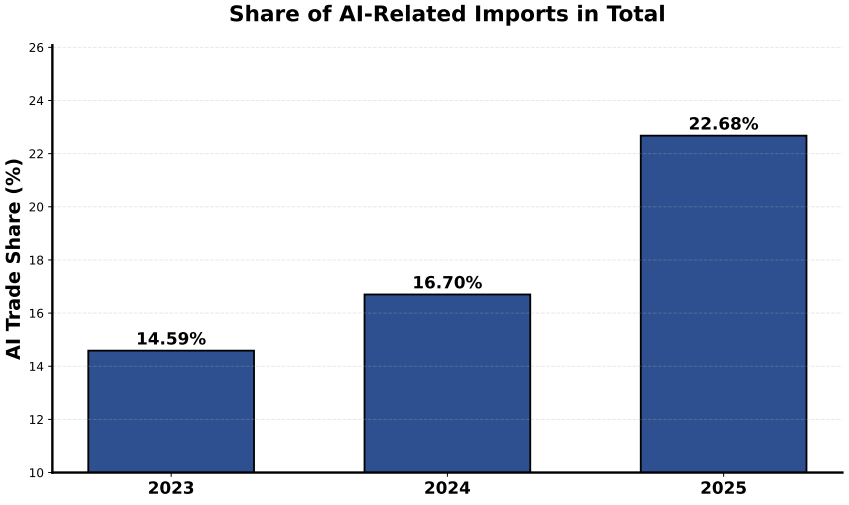
HS10 Code	Description	Category	2025 ($B)	Share (%)	Δ (%)
8471500150	Data processing units	Compute HW	163.6	5.67	+343
8517620090	Telecom transmission equip.	Network	52.9	1.83	+58
8473301180	Printed circuit assemblies	Compute HW	48.9	1.70	+200
8517620020	Switching & routing apparatus	Network	30.8	1.07	+89
8473301140	Memory modules	Compute HW	25.8	0.89	+206
8523510000	SSDs	Compute HW	17.5	0.61	+85
7403110000	Refined copper cathodes	Electrical	16.6	0.58	+150
8507600020	Lithium-ion batteries	Electrical	16.5	0.57	+11
8537109170	Electrical switchgear	Electrical	11.3	0.39	+13
8471804000	Units for ADP machines	Compute HW	11.1	0.39	+937

All ten products show double-digit growth; five show **triple-digit** growth since 2023.

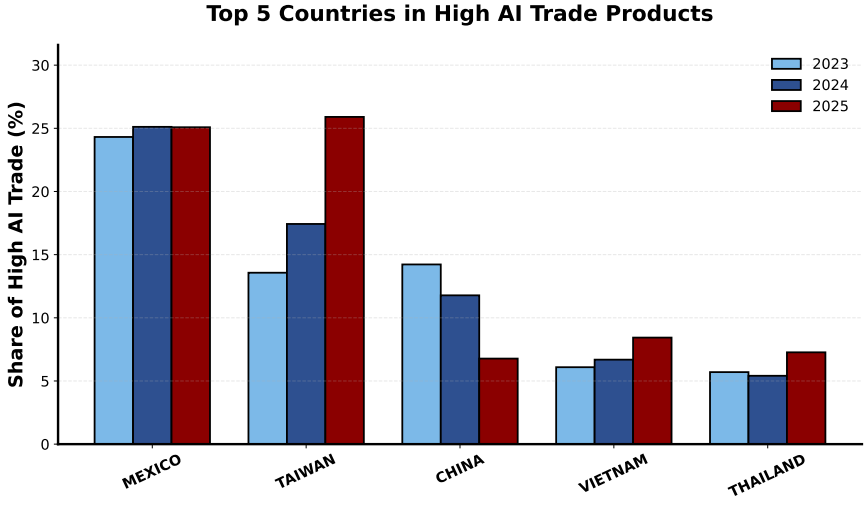
How Much Trade in AI-Related Products? A lot.



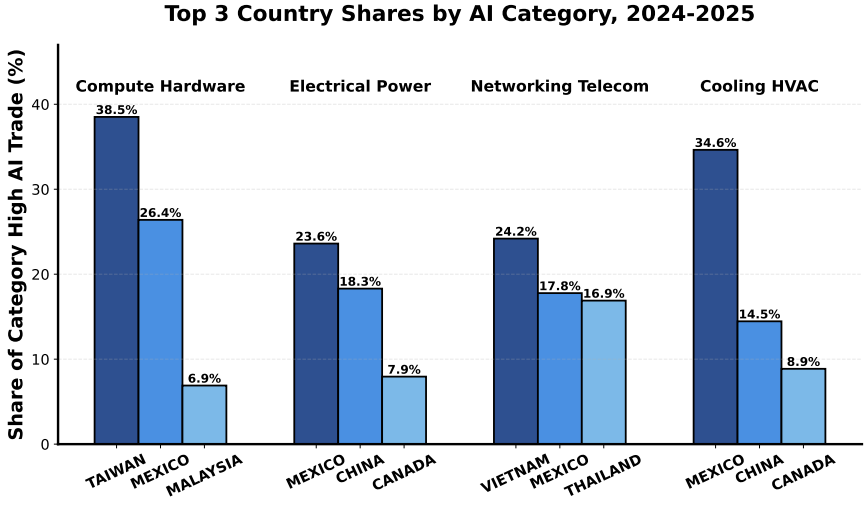
How Much Trade in AI-Related Products? — Share of Imports



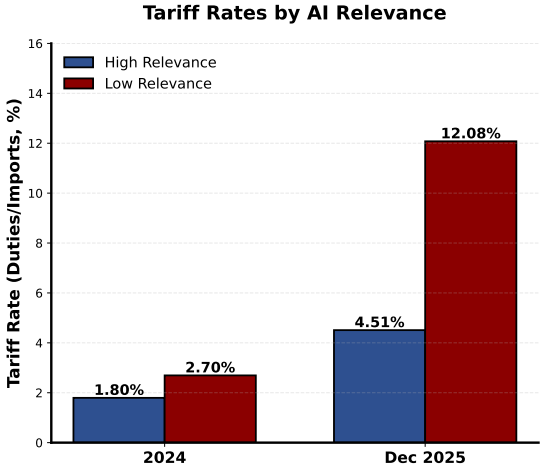
Where Are These Products Coming From? Mexico and Taiwan

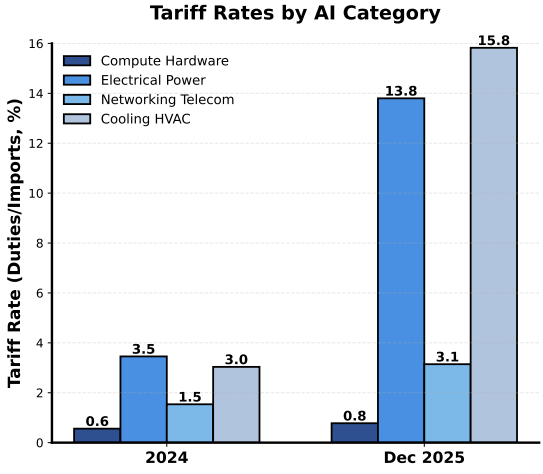


Where Are These Products Coming From? — By Category



How Has Trade Policy Treated AI-Related Products? Lightly.





How Has Trade Policy Treated AI-Related Products? — Exemptions

Exemption List	# HS10 Codes	AI High Trade ($B)	Share (%)
Consumer Electronics (EO 14257)	109	424.5	64.9
Annex II (EO 14257)	720	19.4	3.0
Section 122 Surcharge (Feb 2026)	1,098	445.9	68.2
Any Exemption List	—	**448.6**	**68.6**

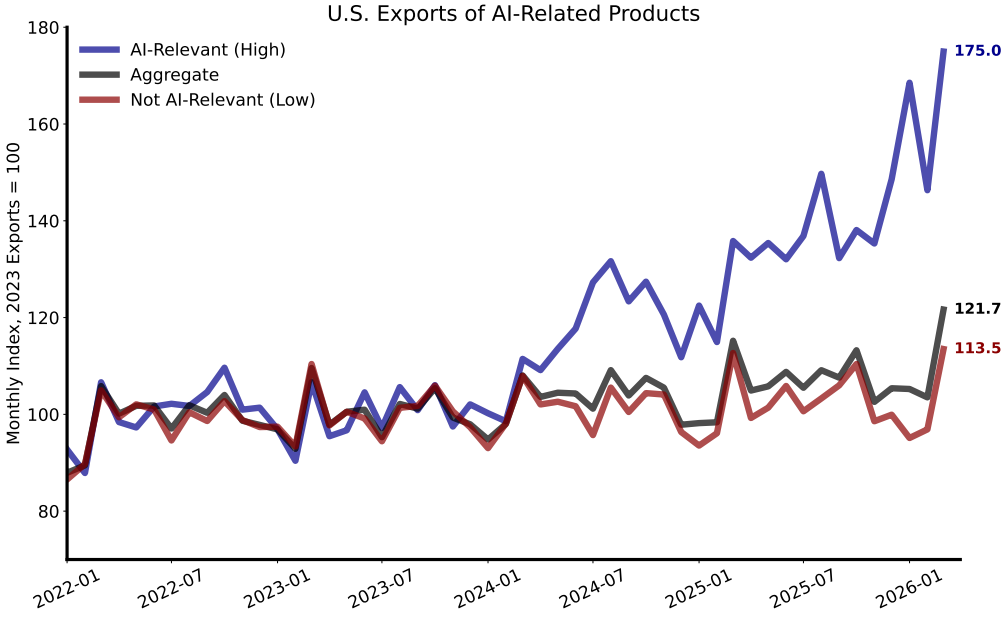
Nearly **69%** of AI-related imports by value fall on at least one exemption list. Compute Hardware is almost entirely shielded — 97% of its $354B in imports is exempt.

U.S. Exports of AI-Related Products

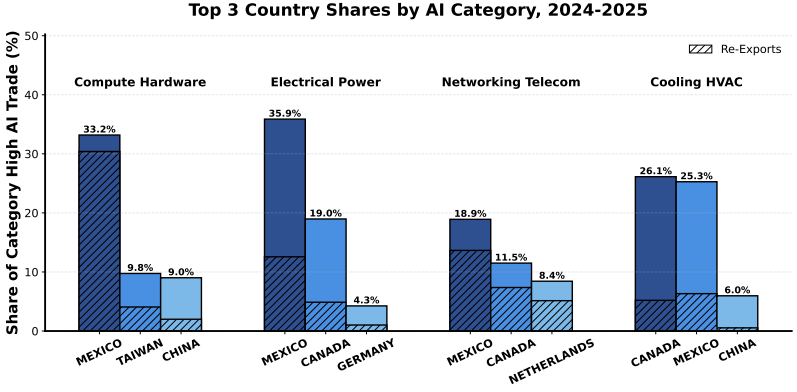
The export side tells a complementary story — U.S. exporters play an important role in the global AI value chain.

Category	# HS10 Codes	2023 ($B)	2025 ($B)	Change (%)
High AI Relevance	**534**	**221.5**	**297.9**	**+34.5**
Compute Hardware	112	99.1	160.6	+62.1
Electrical Power	219	64.5	70.1	+8.8
Networking Telecom	23	29.1	37.7	+29.7
Cooling HVAC	128	24.1	25.0	+3.5
Low AI Relevance	8,278	1,034.8	1,058.6	+2.3
Total Trade	**9,269**	**1,551.9**	**1,648.4**	**+6.2**

U.S. Exports of AI-Related Products



U.S. Exports of AI-Related Products — Destinations



Mexico is an important *destination* for AI-related exports — suggestive that AI materials come into the U.S., go to Mexico, and back.

Does This Matter for the Aggregate? Yes.

A simple accounting exercise:

What would the U.S. goods trade deficit have been if AI-related products had grown the same as non-AI products?

Year	Imports	Exports	Excess AI Imp.	Excess AI Exp.	Net Effect
2023	2,598	1,552	—	—	—
2024	2,792	1,601	66	33	+32
2025	2,884	1,648	265	71	**+194**

Summing Up...

- AI-related products account for **23%** of U.S. imports in 2025.
- Mexico and Taiwan each account for ∼25% of AI trade — Mexico's role is the surprise.
- Effective tariffs on AI products are lower than on other goods — exemption lists cover 69% of AI imports by value.
- The U.S. goods trade deficit would have been $194B smaller absent the AI boom.

Paper, data, and code: https://github.com/mwaugh0328/ai-trade-index